

LEC8 - What is Science?

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The Scientific Method

1. Make observations
2. Ask a question about the observations
 - ↳ Why does — happen?
3. Suggest a hypothesis to explain your observations
4. Make a prediction
 - ↳ Apply hypothesis to something else
5. Perform an experiment
 - ↳ Verify your prediction
6. Make a conclusion / Repeat

Hallmarks of a Scientific Theory

- Based on natural causes
 - ↳ Physics etc.
- Simplicity (Occam's Razor)
 - ↳ As simple as possible while explaining your observations
- Falsifiability
 - ↳ Must be specific and able to be disproven

What is Astrology?

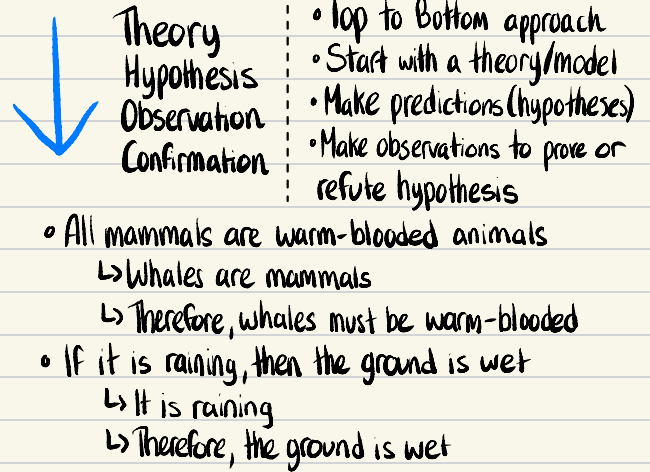
- Astrology developed alongside Astronomy
- Astrology involves using the motion of stars to make predictions about the Earth
- Consistent with random selection

Is Astrology a Scientific Theory?

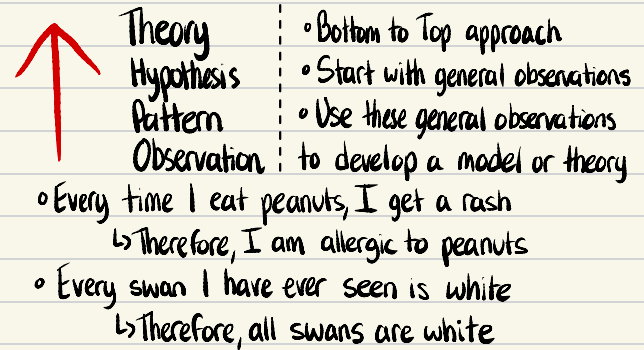
- ✓ ◦ Based on natural phenomenon
 - ↳ Movement of stars
- ✗ ◦ Not simplest explanation
 - ↳ Complicated results
- ✗ ◦ Not easily falsifiable
 - ↳ Results are extremely subjective
 - ↳ Very broadly applicable

Therefore, Astrology is
NOT a scientific theory

DEDUCTIVE REASONING



INDUCTIVE REASONING



Deductive & Inductive Reasoning

- The scientific method is an example of deductive reasoning
- Due to the difficulty of studying distant objects, Astronomy primarily uses inductive reasoning

- Deductive reasoning starts with a theory then uses it to make predictions to prove/refute the theory/model
- Inductive reasoning starts with observations and uses them to make a theory
- Astrology is similar to Astronomy, but the former is not a scientific theory because it does not meet all the requirements to be one